URBAN TRAVEL AND TRANSIT RIDERSHIP

Lecture 6

Public Transport Scenario in India

What is the “Desirable Share of PT”?

- From UT, GoI
- From National Commission
- From MoUD Report
- From RITES Report

Desirable Share of PT

Report on Alternative Systems of Urban Transport, GOI, 1984

City Size	Share of PT trips
<1 million	30%
1 million	35%
1.5 million	40%
3 million	50%
6 million	70%
9 million	75-80%

Desirable Share of PT-2

National Commission on Urbanization - 1987

- In cities with populations exceeding 5 million, the Public Transport share should be 80 %.

Desirable Share of PT - 3

MoUD Report, 1998

City Size	%Share of PT
1 – 5 lakhs	35
5 – 10 lakhs	45
10 – 20 lakhs	55
20 – 50 lakhs	65
> 50 lakhs	75

Desirable Share of PT- 4

RITES Report, 1998

City Size	%Share of PT
1 – 5 lakhs	19
5 – 10 lakhs	26
10 – 20 lakhs	32
20 – 40 lakhs	41
40 – 80 lakhs	53
> 80 lakhs	70

Public Transport Scenario In India

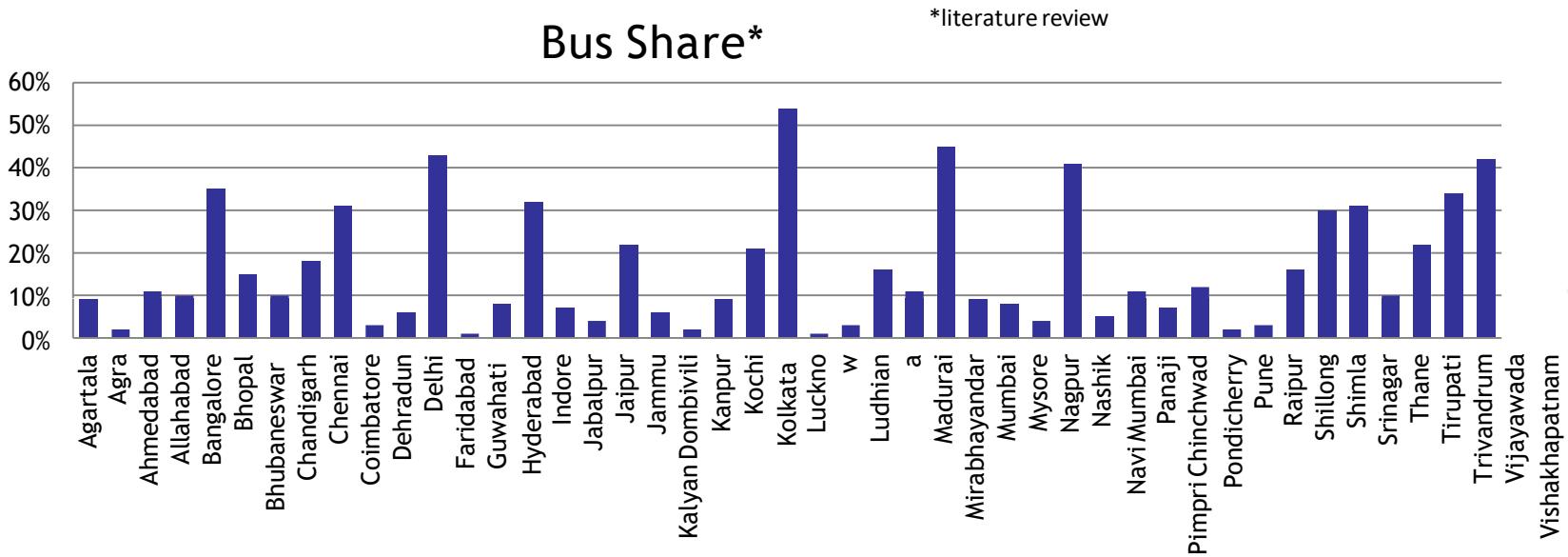
Recommended PT Share City-wise

City	Recommended share	Actual	Gap
Nagpur	65	10	60
Varanasi	55	9	46
Panipat	35	1	34
Ludhiana	55	1	54
Udaypur	35	6	29
Kanpur	65	18	47
Pune	65	29	36
Ahmedabad	65	27	38

Shortage is acute in all classes of cities

Public Transport Scenario In India

PT Share (Bus-Based)



- Kolkata has the highest Bus share (54%)
- Lucknow & Faridabad have the lowest share (1%)
- 65% of the cities have a Bus share of less than 20%

Public Transport Scenario in India

Estimated mode share for future years

Population	2011			2021			2031		
	PT	PV+IPT	NMT	PT	PV+IPT	NMT	PT	PV+IPT	NMT
< 5 lakhs	4	59	36	3	66	31	2	72	26
5 to 10 lakhs	8	42	50	6	51	43	5	58	36
10 to 20 lakhs	12	46	43	10	52	38	9	57	34
20 to 40 lakhs	9	49	42	8	51	41	8	52	40
40 to 80 lakhs	21	45	35	15	51	34	12	54	34
Above 80 lakhs	42	28	30	31	40	29	26	46	28

Source: Study on Traffic & Transportation policies & strategies in urban areas in India – M.O.U.D (2008).

P.T-Public Transport, PV-Personal Vehicles, IPT-Auto rickshaws, NMT-Non motorized transport including walking and cycling

Public Transport Scenario in India

PT Bus-Based

- Bus services (both inter & intra city) comprise about 90% of Public Transport
- Modal share of buses varies from 0% in some small cities to 40% or more in larger cities (Bangalore & Delhi)
- Modal share of buses in Delhi diminished from 60% in the 1990s to 40% in 2008 (source: DIMTS 2009)
- People spend nearly 10% to 15% of their income on Transport

Public Transport Scenario in India

"Conclusions" from current scenario

- Increasing private vehicles
- Decreasing public transport vehicles

Why people use private modes?

- Social and economical influence
- Personal and family transport
- Unlimited commercial versatility
- Door to door service
- Flexible routing
- Indispensable for quality of life

… let’s try to answer these questions

Why is the Public Transport share declining ?

Why don’t people like to use Public Transport ?

Why People Don't Use PT

- Inconvenience in terms of location of bus stops / stations and low frequency of services
- Failure to service key Origins – Destinations
- Lack of seamless travel
- Fear of crime at stations and within buses
- Lack of safety in terms of accidents and roadworthiness of vehicles / buses
- Lack of passenger information

Why People Don’t Use PT - 2

- Service is slower than private vehicles and buses make frequent stops
- Overloading of vehicles makes rides uncomfortable
- Public transport is more expensive than 2-wheelers
- Poor quality infrastructure (shelters, bus stations, vehicles)
- Low status of public transit services

Barriers for Improvement of PT

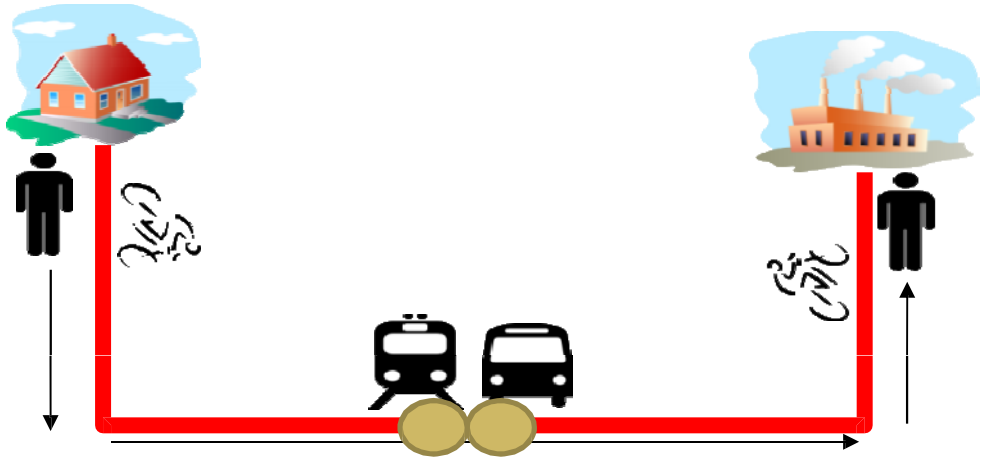
- Lack of political will
- Governance
- Opposition from key stakeholders (operators, motorists)
- Political and institutional inertia
- Institutional biases

- Lack of information
- Poor institutional capacity
- Inadequate technical capacity
- Insufficient funding and financing
- Geographical/physical limitations

… let’s understand the Urban Transport problems

and hence the need for Public Transport

So far…



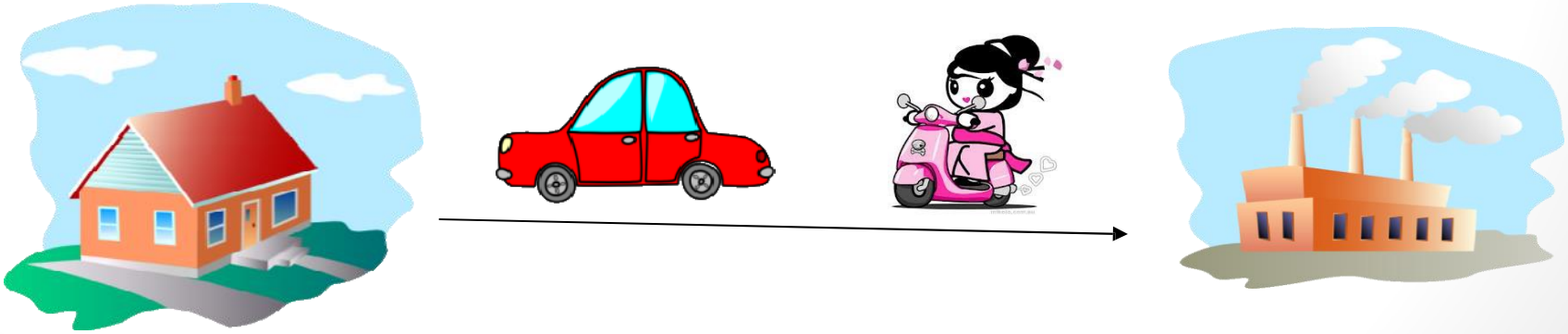
Source: GIZ

Lack of….
Information, Integration, Design, Priority, Security, Options, Fares, Coverage, Frequency, …

...instead of all of the above

Why can't I just drive to work ?

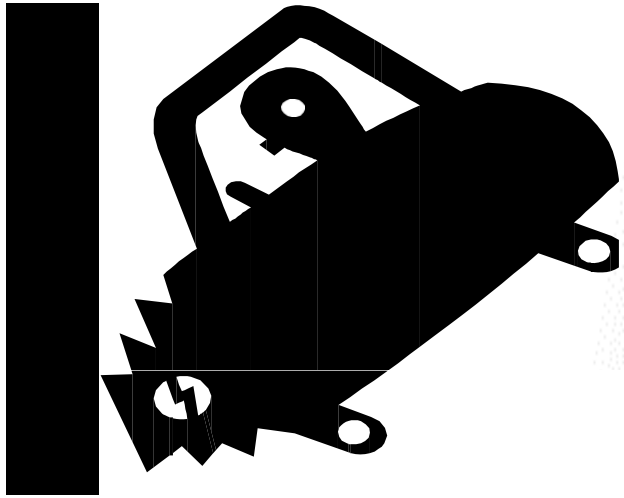
- It's easy
- Starts from my door step
- It's faster
- I don't pay for a ticket



Source: GIZ

Costs of Owning a Private Vehicle

- Health costs: self and others
- Environmental costs
- Livability in the city
- Fuel and insurance costs
- Maintenance costs
- Congestion costs
- Opportunity cost of space



Source: GIZ

HOW FAR IS SOCIETY PREPARED TO GO WITH TRAFFIC PROBLEMS?

Urban Transport Problems

Accidents

Traffic Jams

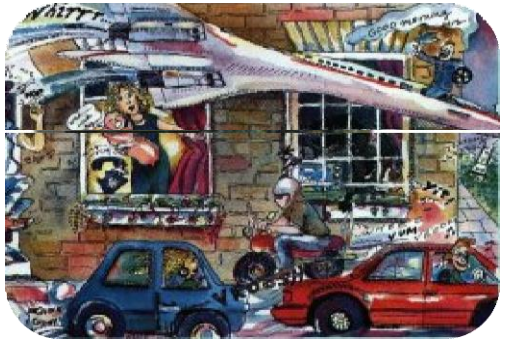
Frustrations

Source: World Wide Web

Urban Transport Problems - 2

Parking Difficulties

Noise



Pollution/Emissions

Source: World Wide Web

Need for Good Public Transport

Four Reasons

- To Counter ----- Congestion on Roads
- To Minimize ----- Energy Consumption
- To Ensure ------- Healthy Environment
- To Improve ------ Safety for all

Need for Good Public Transport - 2

Totransport 10,000 people for one - kilometer						
Parameters	**Car**	**Minibus**	**Regul ar Bus**	**Heav y Bus**	**Articulat ed Bus**	**Bi- articulated Bus**
Persons per vehicle	5	25	80	105	180	270
Vehicles needed	2000	400	125	95	55	37
Area occupied (sq. m)	48,000	8,800	3,000	3,260	2,600	2,370
Fuel Consumption (liters)	400	120	40	30	31	34

Source: Transport and Environment Report, Department of Ecology and Environment, Govt. of Karnataka

Benefits of PublicTransport

- Time savings – benefit to transit users
- Fuel savings from transport operations
- Air quality improvement (reduced emissions)
- Green house gas reduction
- Noise and vibration reduction
- Other environmental improvement (reduced solid and liquid waste, reduced impact on flora and fauna)

Benefits of Public Transport - 2

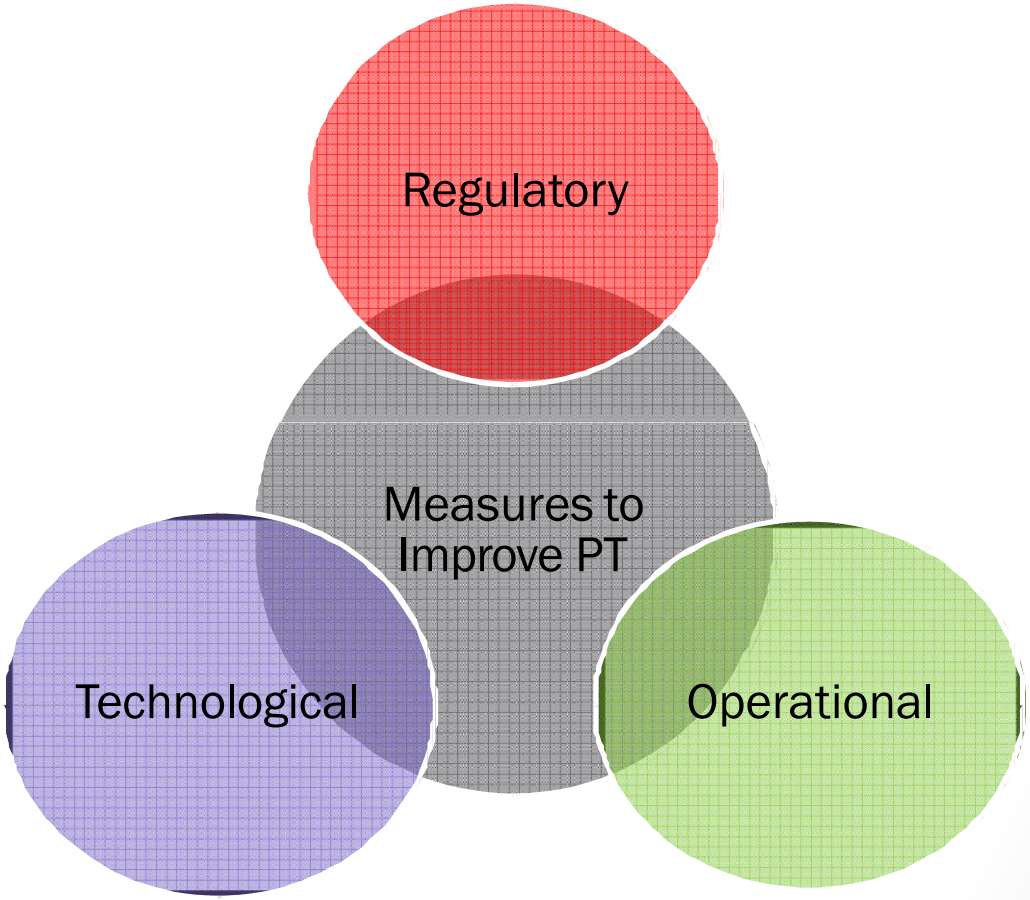
- Public Transport system employment
- Commercial sector
- Social benefits
- City image
- Urban form
- Political
- Amenity benefits to passengers: comfort, prestige

… what are the measures to improve Public Transport ?

Measures to Improve Public Transport System

Three main measures…

- Regulatory
- Operational
- Technological



Action Plans to Solve Traffic Problems

A package of actions is needed

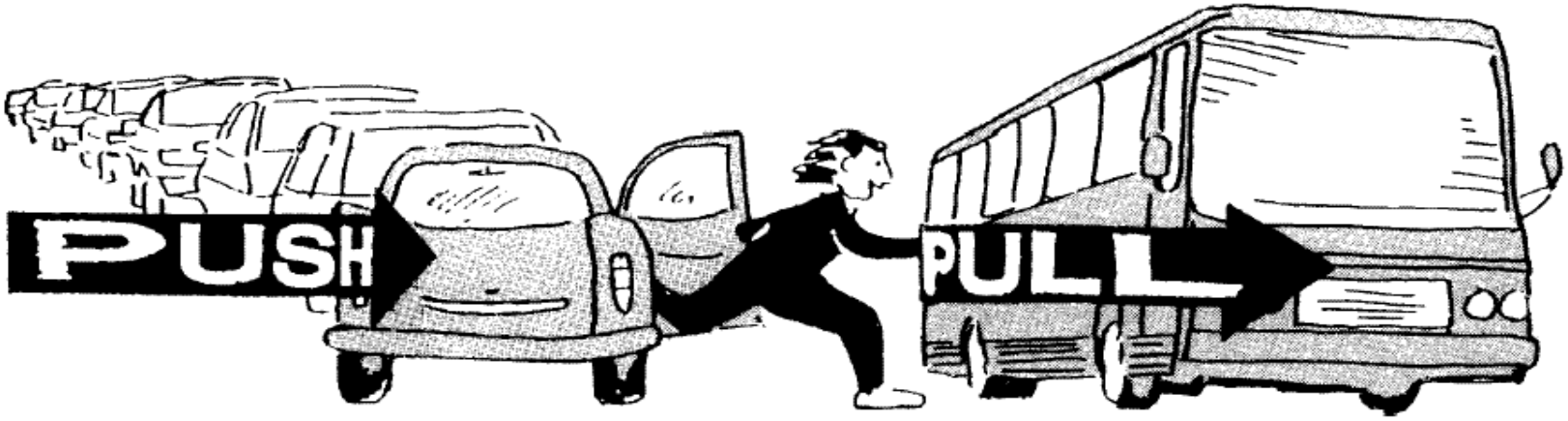
- Restraint on private cars
- Encouragement to Public Transport
- Parking controls in central areas
- Traffic Management
- Bus-only lanes and links
- Computer controlled traffic signal systems

Tackling the Problem

The Push and Pull Approach (strong Focus on TDM measures required)

Measures with push-effects
Area-wide parking management, parking space restrictions in zoning ordinances, car limited zones, permanent or time-of-day car bans, congestion management, speed reductions, road pricing...

Measures with pull-effects
Priority for buses and trams, high service frequency, passenger friendly stops and surroundings, more comfort, park-and-ride, bike-and-ride..., area-wide cycle-networks, attractive pedestrian connections...



Measures with push- and pull-effects
Redistribution of carriageway space to provide cycle lanes, broader sidewalks, planting strips, bus lanes..., redistribution of time-cycles at traffic lights in favour of public transport and non-motorized modes, public-awareness-concepts, citizens' participation and marketing, enforcement and penalizing...

Source: Müller, P., Schleicher-Jester, F., Schmidt, M.-P. & Topp, H.H. (1992): Konzepte flächenhafter Verkehrsberuhigung in 16 Städten", Grüne Reihe des Fachgebiets Verkehrswesen der Universität Kaiserslautern No. 24.

4 steps regarding policy framework for public transport:

1. Clearly define the role of government in urban transport:

- providing policy directions
- rewarding good practices financially
- adopting performance benchmarking/measurements
- guiding municipal financial reform

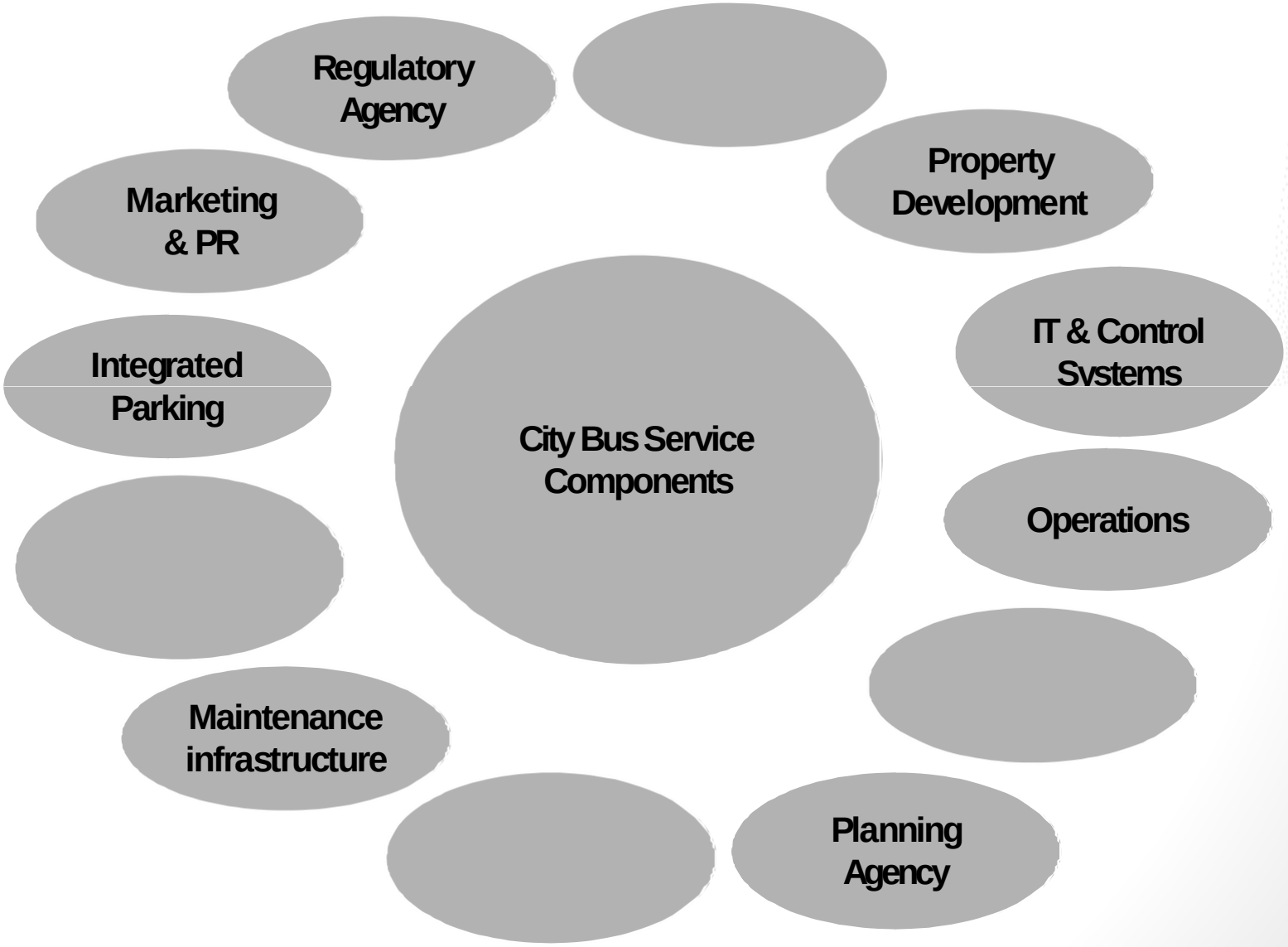
2.Develop accountabilty procedures and promote public participation

3.Strengthen institutional and technical capacity for strategic planning and monitoring

4. Integrate urban transport planning and operations for the entire metropolitain area

Source: GIZ

City Bus Service Components



Bus Service and Operations Design

- Travel demand assessment
- Route planning
- Operations plan – schedule, frequency, etc.
- Fare structure and collection system design
- Costing and financial analysis
- ITS and control systems design
- Marketing strategy, branding and promotion
- Institutional options

Route and Service planning for city bus service……

Route Planning

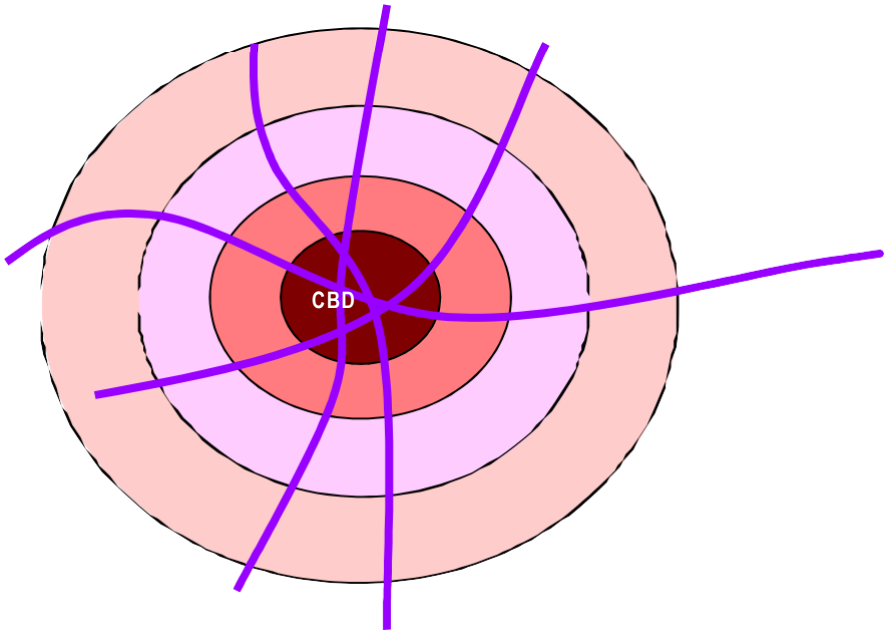
Surveys to be conducted

- Road Network Inventory
- Classified Traffic Volume Count
- Origin and Destination Survey
- Speed and Delay survey
- Occupancy survey
- Bus Boarding-Alighting Survey
- Willingness to Pay Survey

Route Planning -3

Types of Bus Route Networks

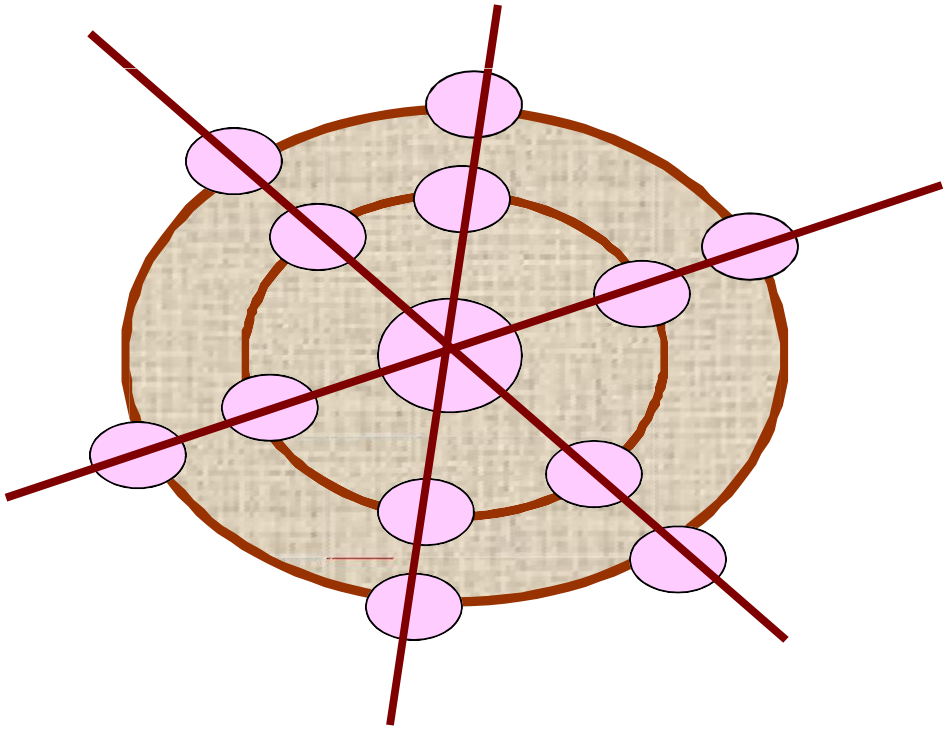
- **RADIAL** - Suitable for cities with strong central core around which the development has taken place. Population density reduces from CBD to fringes



Route Planning -4

Types of Bus Route Networks

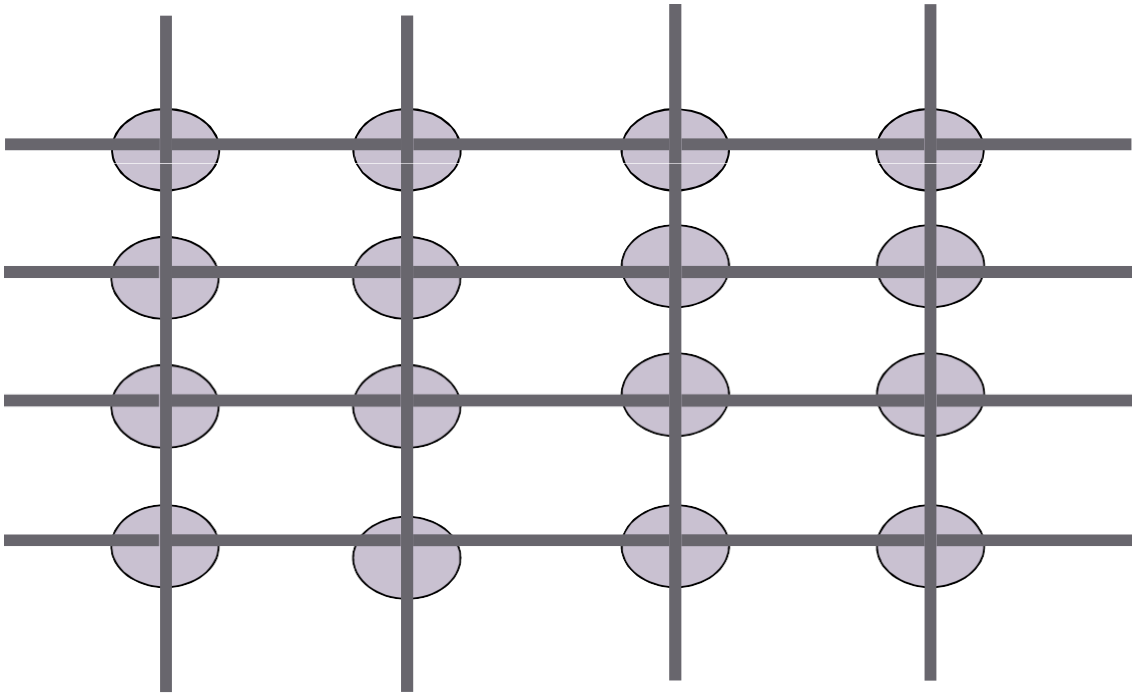
- **RADIAL AND CIRCULAR** - Suitable for cities where the activity centers are developed along radial corridors



Route Planning -5

Types of Bus Route Networks

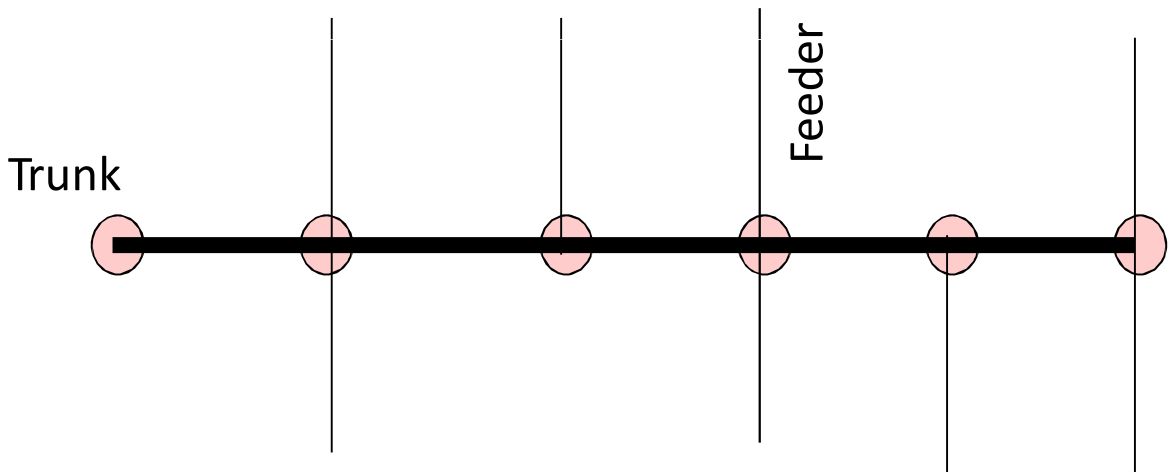
- **GRID -** Suitable for cities having multiple activity centers spread uniformly throughout



Route Planning -6

Types of Bus Route Networks

- **TRUNK AND FEEDER** - Suitable for cities that have evolved linearly along a major corridor and the activity centers are spread parallel along the corridor.



Route Rationalization Criteria

- Physical Criteria
  - Inventory of road and land use
- Operational Criteria
  - analysis of operational performance
- Behavioral Criteria
  - change of traffic pattern
  - introduction of other modes
  - crew behavior

Rationalization of Existing Bus Routes

S. No.	Stages	Description
1	Stage I	Analysis of existing bus ridership - elimination/ modification of overlapping routes and/or routes with lesser demand
2	Stage II	Elimination of routes parallel to proposed bus network
3	Stage III	Elimination/Merging of shorter bus routes
4	Stage IV	Comparison of bus service network with major auto routes in the city and addition of routes which are currently unserved by the bus network
5	Stage V	Comparison of 'Origin- Destination survey data' with existing bus/ auto routes and addition of routes/ connection for OD pairs having higher number of trips.

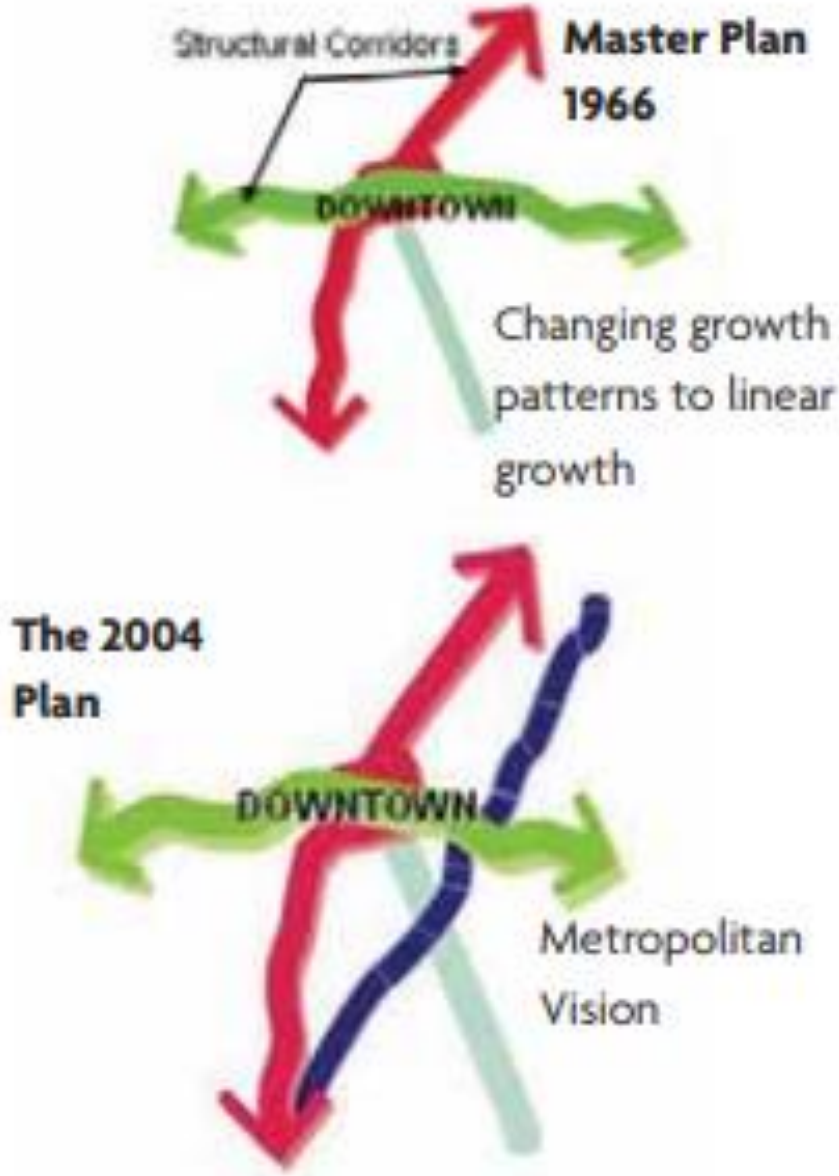


Figure 3.3 Urban Growth Axes in Curitiba

Source: IPPUC (2009a).

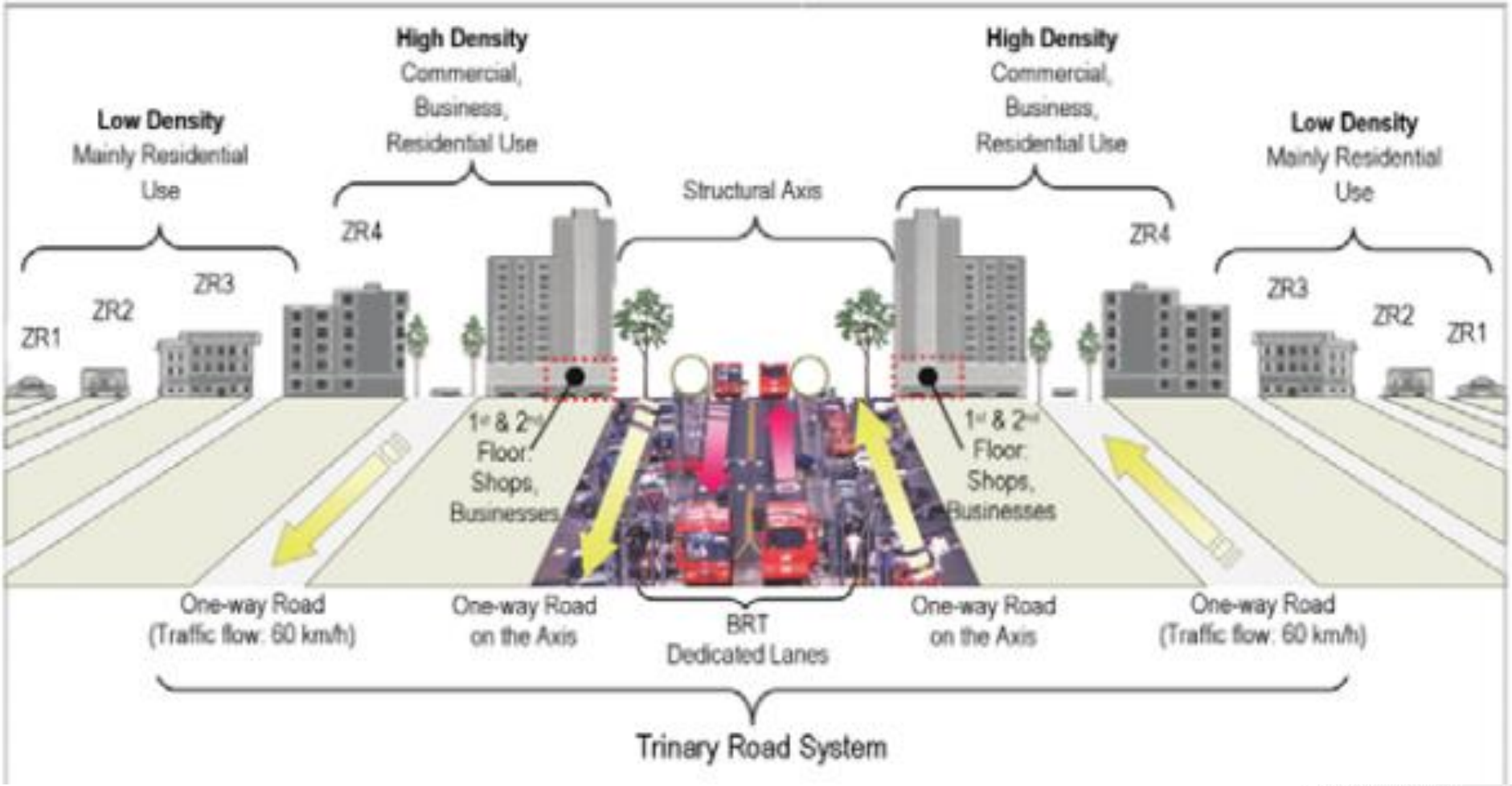


Figure 3.7 The Trinary Road System in Curitiba

Source: Author compilation (Hinako Maruyama) based on IPPUC (2009a), Hattori (2004), and pictures supplied by IPPUC.
Note: km/h = kilometers per hour.